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ORIGINAL RESEARCH

Enhanced hand-gesture recognition by improved beetle swarm optimized probabilistic neural network for human–computer interaction

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Abstract

The dynamic hand gesture recognition is a new research area due to the emerging nature of pervasive somatosensory devices, which has been gained more attention and been broadly utilized in human–computer interaction (HCI). This paper tactics to develop an intelligent hand gesture recognition model through considering the use of hand gestures for HCI. The user-friendly, less intrusive, intuitive, and more natural HCI for controlling an application by executing hand gestures are presented. The overall models cover diverse steps like data collection, pre-processing, segmentation, feature extraction, and recognition. The hand gestures videos from different benchmark sources are collected, and further, the pre-processing is performed by median filtering. Further, the goal of the segmentation is to evaluate the temporal hand trajectories from the detected hand poses, which is carried out through the adaptive region-based active contour (ARAC) method based on the meta-heuristic basis by the opposition strategic velocity updated beetle swarm optimization (OSV-BSO). Feature extraction is the next step, which is done by combining a set of features like oriented FAST and rotated BRIEF (ORB), histogram of oriented gradients (HOG), and regionprops, which are further given to the principal component analysis (PCA) for dimensionality reduction. With these relevant features, the recognition phase is accomplished by the optimized probabilistic neural network (PNN). To improvise the existing performance of PNN as a deep learning model, the weights in training are tuned based on the meta-heuristic basis by the OSV-BSO. The objective model of the optimized PNN is to minimize the error between the desired and actual outcomes. Finally, the designed "gesture recognition approach" makes the HCI process more user-specific and intuitive, which is proved by comparing it over the existing approaches.

Keywords Dynamic hand gesture recognition · Human–computer interaction · Adaptive region-based active contour · Optimized probabilistic neural network · Opposition strategic velocity updated beetle swarm optimization

1 Introduction

HCI based on "hand gestures" plays a major role in diverse modalities as humans generally lay on their hands in transmission or for interacting with their environment. The existing "hand-gesture-based approaches" are different from other methods by offering a common way of communication and interaction (Köpüklü et al. 2020; Hu et al. 2019; Maqueda et al. 2016). Numerous studies analyze "gesture-based interaction" approaches, their challenges, and advanced ways for enhancing its efficiency (Bao et al. 2017; Liu et al. 2018; Chen et al. 2019). In recent years, "hand-gesture recognition" models by considering a computer vision have experienced a gradual development in popularity because of the broader range of probable applications in the area of HCI like medical systems (Ameur et al. 2020) video games (Rautaray and Agrawal 2015), and multimedia application control (Plouffe and Cretu 2016). The interfaces in hand gestures like more natural, less intrusive, friendly, and intuitive are considered to each user than conventional HCI devices. The usage of mouse and keyboard can be more helpful to some other applications, in which the major advantage is hand-based interfaces. However, the majority of the customer devices are inserted with color cameras that motivate the development of HCI models with the design of color-based methods and

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hand-gesture recognition (Li et al. 2020; Su 2000; Deng 2020). Spectral clustering (SC) has been widely applied to various computer vision tasks, where the key is to construct a robust affinity matrix for data partitioning (Zhihui et al. 2018). A rank-constrained Spectral Clustering (SC) is proposed with a flexible embedding framework. Specifically, an adaptive probabilistic neighborhood learning process is employed to recover the block-diagonal affinity matrix of an ideal graph. Meanwhile, a flexible embedding scheme is learned to unravel the intrinsic cluster structure in low-dimensional subspace, where the irrelevant information and noise in high-dimensional data have been effectively suppressed (Li et al. 2018). A new zero-shot event detection approach is proposed, which exploits the semantic correlation between an event and concepts. Based on the concept detectors pre-trained from external sources, this method learns the semantic correlation from the concept vocabulary and emphasizes the most related concepts for the zero-shot event detection. Particularly, a novel Event-Adaptive Concept Integration algorithm (E-ACI) is introduced to estimate the effectiveness of semantically related concepts by assigning different weights to them (Li et al. 2019).

Gesture recognition is a broad task for integrating diverse criteria of computer science like machine learning, pattern recognition, motion analysis, and motion modeling (Duan et al. 2020). In the 1990s, numerous hand gesture recognition methods have been developed that are categorized into two types such as “glove-based or vision-based”, which is divided based on the motion capture technique. Glove-based approaches are generally offered more consistent motion data and remove the requirement to middle-tier software for capturing the hand postures and positions (Skaria et al. 2019). Though, it needs the user to wear position trackers and “cumbersome data gloves” that generally hold some connection cables, which minimizes the usefulness and intuitiveness of such approaches and makes it costly (Ohn-Bar and Trivedi 2014). Vision-based methods depend on image processing approaches for extracting posture and motion trajectory information (Lu et al. 2014; Köpüklü et al. 2020). Its achievement is majorly based on the image analysis approaches that are perceptive to the environmental aspects like changes in illumination and may lose fine details because of the finger and hand occlusion (Tu and Huang 2019).

The major contributions of the designed model are given here.

- The hand gestures videos are collected from different benchmark sources. A new intelligent hand gesture recognition model is proposed through an improved segmentation approach and optimized deep learning approach with the aid of a new meta-heuristic-based technique for HCI to get better-classified outcomes.
- To implement a novel blood cell segmentation model through a new segmentation approach called ARAC method by optimizing the parameters of region-based active contour technique by the OSV-BSO. The efficiency is verified by considering the multi-objective functions concerning with maximization of entropy and minimization of variance to get accurate segmented results.
- To offer accurate classified hand gestures using optimized PNN to maximize the precision to get promising results. It is performed through the OSV-BSO.
- To evaluate the efficiency of the suggested hand gesture classification model for HCI via evaluating some standard measures by comparing with standard techniques.

The remaining sections of the designed model are given here. Section 2 analyzes the related works. Section 3 specifies the improved segmentation and recognition model for “hand gesture recognition”. Section 4 explains the “hand gesture recognition” by ARAC. Section 5 indicates the proposed OSV-BSO for enhancing segmentation and classification. Section 6 evaluates the results. Section 7 gives a conclusion.

2 Literature survey

2.1 Related works

In Yu et al. (2019) have proposed the semi-supervised model named adaptive semi-supervised feature selection for cross-modal retrieval. The semantic regression was utilized to strengthen the neighboring relationship between the data with the same semantic framework. The correlation between heterogeneous data can be optimized via keeping the pairwise closeness when learning the common latent space. The graph-based constraint was adopted to predict accurate labels for unlabeled data, and it can also keep the geometric structure consistency between the label space and the feature space of heterogeneous data in the common latent space. Finally, an efficient joint optimization algorithm was proposed to update the mapping matrices and the label matrix for unlabeled data simultaneously and iteratively. Experiment results have demonstrated that our method performed better than the state-of-the-art methods for three datasets.

In Zhang et al. (2020) have proposed a novel deep multimodal network with a top-k ranking loss to mitigate the data ambiguity problem. With this strategy, query results do not have been penalized unless the index of ground truth was outside the range of top-k query results. Considering the non-smoothness and non-convexity of the initial top-k ranking loss, a tight convex upper bound was exploited to approximate the loss and then utilize the traditional back-propagation algorithm to optimize the deep multi-modal network. Empirical results on metrics have shown that our method achieved comparable performance in comparison to state-of-the-art methods.

In Zhou et al. (2020) have proposed a multi-feature fusion with an adaptive graph learning model for unsupervised (Re-Identification) Re-ID. This model aimed to negotiate comprehensive assessment on the consistent graph structure of pedestrians with the help of special information of feature descriptors. Specifically, this proposed model was incorporated multi-feature dictionary learning and adaptive multi-feature graph learning into a unified learning model such that the learned dictionaries were discriminative and the subsequent graph structure learning was accurate. An alternating optimization algorithm with proved convergence was developed to solve the final optimization objective. Extensive experiments have demonstrated the superiority and effectiveness of the proposed method for four benchmark data sets.

In Luo et al. (2018a) have exploited a novel semisupervised feature selection method from a new perspective. This proposed model was the instances with similar labels should have a larger probability of being neighbors. Instead of using a predetermined similarity graph, this proposed model was incorporated the exploration of the local structure into the procedure of joint feature selection to learn the optimal graph simultaneously. Moreover, an adaptive loss function was exploited to measure the label fitness, which significantly enhances the model's robustness to videos with a small or substantial loss. An efficient alternating optimization algorithm has been proposed to solve the proposed challenging problem, together with analyses on its convergence and computational complexity in theory. Finally, extensive experimental results have illustrated the effectiveness and superiority of the proposed approach on video semantic recognition-related tasks for benchmark datasets.

In Ren et al. (2021) have reviewed a comprehensive and systematic survey on the NAS was essential. Previously related surveys have begun to classify existing work mainly based on the key components of NAS: search space, search strategy, and evaluation strategy. While this classification method was more intuitive, it is difficult for readers to grasp the challenges and the landmark work involved. Therefore, this survey has provided a new perspective: beginning with an overview of the characteristics of the earliest NAS algorithms, summarizing the problems in these early NAS algorithms, and then providing solutions for subsequent related research work.

In Luo et al. (2018b) have characterized the intrinsic local structure by an adaptive reconstruction graph and simultaneously considered its multiconnected-components (multicluster) structure by imposing a rank constraint on the corresponding Laplacian matrix. To achieve a desirable feature subset, it has learned the optimal reconstruction graph and selective matrix simultaneously, instead of using a predetermined graph. An efficient alternative optimization algorithm has been exploited to solve the proposed challenging problem, together with the theoretical analyses on its convergence and computational complexity. Finally, extensive experiments on clustering tasks were conducted to verify the effectiveness and superiority of the proposed unsupervised feature selection algorithm for several benchmark data sets.

In Chang et al. (2016) have proposed a novel Compound Rank-K Projection (CRP) algorithm for bilinear analysis. The CRP deals with matrices directly without transforming them into vectors, and it, therefore, preserves the correlations within the matrix and decreases the computation complexity. Different from the existing 2-D discriminant analysis algorithms, objective function values of CRP increase monotonically. In addition, the CRP utilizes multiple rank-k projection models to enable a larger search space, in which the optimal solution can be found. In this way, the discriminant ability was enhanced. Experimental results have shown that the performance of our proposed CRP performed better than other algorithms in terms of classification accuracy.

2.2 Problem statement

In industries, hand gesture-based HCI models are required largely. Therefore, developing a robust hand gesture recognition model will remain a complex issue. Finite-state machine-based recognition (Kılıboz and Güdükbay 2015) improves the accuracy, and it can be implemented in real-time applications. The proposed model cannot be extended for the third dimension using the depth (z-coordinate), and it is also restricted for dynamic posture recognition. VS-LBP and SVM (Maqueda et al. 2015) are efficient based on computational efforts. It attains high recognition accuracy. The performance of this model is affected. GA-ELM (Qi et al. 2019) improves the accuracy of gesture recognition. Long-term gesture recognition can be possible. Though, it requires more training time. Motion-Feature-Warping Algorithm and Shape-Feature-Warping Algorithm (Zhu and Xu 2002) eliminate the temporal variations and enhance the recognition rate. However, it has computational complexity. HMM-PSO (Liu and Huai 2019) has a better recognition rate and enhances a better interactive experience. On the other hand, the gesture cannot be recognized accurately. The performance of this model is not efficient. Component labeling algorithm (Tsai et al. 2020) achieves the low cost and lightweight model. It obtains better accuracy. There is some interference in the model. Multi-view feature learning (Song et al. 2020) can be optimizing the multimodal features globally and locally. It attains better recognition accuracy in less time. Some of the recognition of hand gestures is difficult by using this method. Ensemble learning (Yadav and Bhattacharya 2016) improves classification accuracy. It can enable to evaluate the dynamic gestures. However, this model is limited to hand gestures of scroll up or down depending. These limitations are considered for developing a new Hand-Gesture Recognition model for HCI. From the past researchers, it is complex to convince and clear the differentiation among recently stated gesture recognition models due to the two models are aimed for the same application. Therefore, the difficulties of each model also differ regarding their command set of gestures, system settings, assumptions about the background, etc. There is also a shortage of standard databases as compared with diverse gesture recognition models.

3 Improved segmentation and recognition model for hand gesture recognition

3.1 Proposed architecture for hand gesture recognition

The recognition of hand gestures is important to factor for HCI. The gesture-based recognition models have limitations for using machine learning due to the lack of meaningful and rich datasets. In recent decades, diverse deep learning-based hand gesture recognition models have been developed that have more challenges. The classifiers like SVM and Naive Bayes (NB) are complex to process a huge number of datasets with a requirement of more training samples. NN with gesture classification (Murthy et al. 2021) is a more time-consuming model due to the increase in the number of training data. Fuzzy C-Means clustering (FCM) for gesture recognition is affected because of the false object extraction when the objects are bigger than the hand. Correspondingly, the hand images with poor camera quality impact the performance of the gesture recognition model. Similarly, the segmentation performance is also reduced by the lighting conditions of images. The most significant segmentation technique for gesture recognition is HMM tools. However, this method takes more computational time. Therefore, there is a need to propose a better segmentation and classification approach for increasing the performance of the "proposed hand gesture recognition" model. It is depicted in Fig. 1.

The "proposed hand gesture recognition" in HCI is proposed by applying a new adaptive region-based active contour segmentation and PNN classification. The designed architecture hand gesture recognition model undergoes the

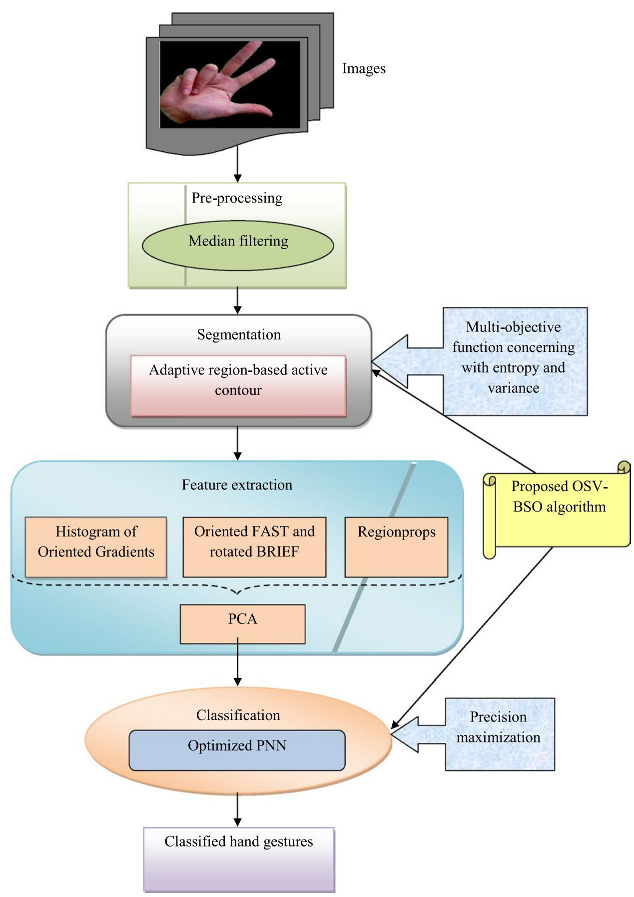

Fig. 1 "proposed hand gesture recognition" model

following steps: (a)data collection, (b)pre-processing, (c) segmentation, (d)feature extraction, and (e)recognition. This model collects the hand gesture images from the standard dataset to process. The collected images are given to the pre-processing phase for removing the noise from the images, which is done by a median filtering approach that intends to improve the results by detecting the edges in an image. It efficiently eliminates the noise without affecting the edges of an image. The pre-processed images are segmented using ARAC, where the constants like ς and ϑ a maximum number of iteration J of the region-based active contour are optimized (Swamy et al. 2013) using the OSV-BSO algorithm. This segmentation process is a more robust approach for improving the convergence behavior by segmenting the regions. The segmentation considers the multi-objective function concerning with maximization of entropy and minimization of variance. The segmented images are further given to the feature extraction phase for extracting the significant features. It is performed by techniques such as HOG, ORB, and region props. Generally, feature extraction aims to reduce the number of resources for describing a huge number of the dataset. All the features are combined and dimensionality reduced by applying the PCA technique. The dimensionality reduced PCA (Huan et al. 2016) features are classified by using PNN, where the weights of PNN are optimized using the OSV-BSO algorithm. Finally, the classified hand gestures are attained by solving the precision maximization.

3.2 Dataset description

The "proposed hand gesture recognition" model has collected the input images from two datasets.

Dataset 1: It is taken from "https://www.kaggle.com/ayuraj/american-sign-language-dataset: accessed date: 2021-01-11". This dataset consists of 25.3 k images; it also includes images with sign language for alphabets with numbers. Thus, it has 36 columns, where each column includes a varied number of images.

Dataset 2: It is taken for experimentation with two datasets. It is one of the hand gesture recognition datasets for alphabets, which includes images with sign language for alphabets and gathered from "https://www.kaggle.com/muhammadkhalid/sign-language-for-alphabets: access date: 2021–01-11". It has 27 classes and the total number of images are taken as 40,500, where each class includes 1500 images. The sign language for numbers is gathered from standard Hand gesture recognition datasets for numbers from the source of "https://www.kaggle.com/muhammadkhalid/sign-language-for-numbers/vaersion/1: access date: 2021-01-11". It has a total number of classes of 11 and a total number of images as 16,500, where each class includes 1500

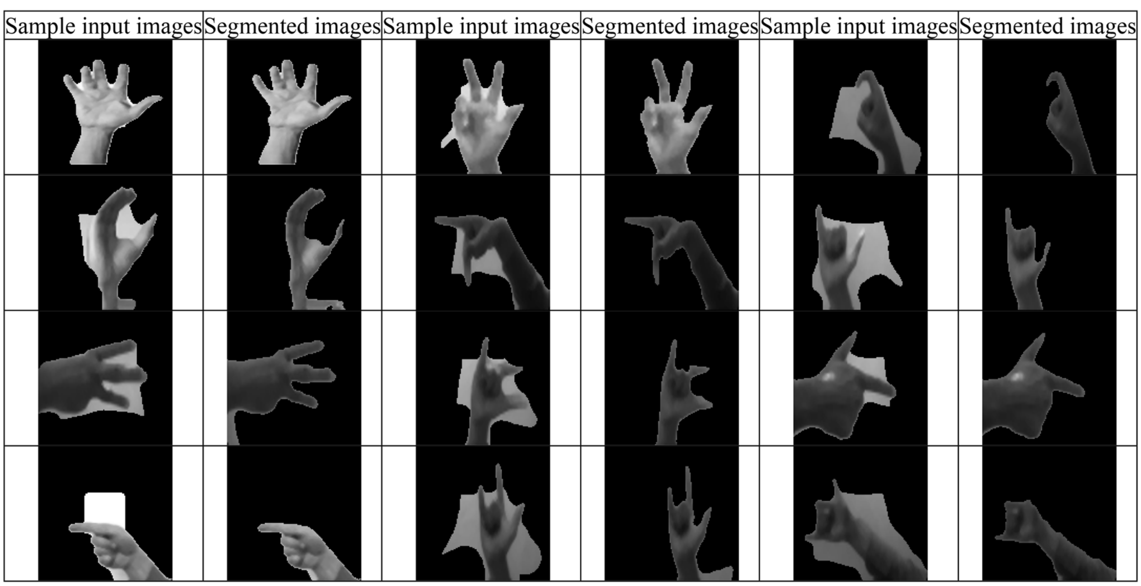

Fig. 2 Simulation results for the "proposed hand gesture recognition" model

images. Thus, dataset 2 includes a total of 57,000 images for experimentation.

3.3 Segmentation results

The sample simulation images of the "proposed hand gesture recognition" model through ARAC with the OSV-BSO algorithm are given in Fig. 2.

3.4 Objective model on segmentation and classification

The "proposed hand gesture recognition" model considers two objective functions that ensure better segmentation and recognition accuracy. The primary objective of the proposed model is to minimize the variance and maximize the entropy of images during ARAC-based segmentation, which is given in Eq. (1):

$$FF_1 = \underset{\{\varsigma,\vartheta,J\}}{\arg\min}\left(\frac{1}{Ent} + Var\right). \tag{1}$$

Here, the parameter α and a maximum number of iteration J of the region-based active contour are optimized using the OSV-BSO algorithm. The terms Ent and Var denotes the entropy and variance of images. The entropy is also considered as average information, which is described as the "measure of the degree of randomness in the image". The Shannon entropy is formulated in Eq. (2):

$$Ent(xd) = -\sum_{ij=1}^{ni} pe\left(xd_{ij}\right)\log_2\left[pe\left(xd_{ij}\right)\right]. \tag{2}$$

Here, the discrete random variable is termed as xd, where $xd = \left\{xd_{ij}, xd_2, \cdots, xd_{ni}\right\}$, the probability of an event xd_{ij} is specified as $pe\left(xd_{ij}\right)$, in which $pe \in [0, 1]$. The variance of the image is formulated in Eq. (3):

$$Var = \frac{1}{NI}\sum_{ij=1}^{NI}\left(xd_{ij} - \overline{xd}\right)^2. \tag{3}$$

In Eq. (3), the term $\overline{xd}$ is mean and xd_{ij} indicates the data vector. The second objective of the "proposed hand gesture recognition" model is the maximization of precision that is shown in Eq. (4):

$$FF_2 = \underset{\{WE_w\}}{\arg\min}\left(\frac{1}{pre}\right). \tag{4}$$

Here, the precision is termed as pre, which is "the ratio of positive observations that are predicted exactly to the total number of observations that are positively predicted" as given in Eq. (5):

$$pre = \frac{t^{pos}}{t^{pos} + f^{pos}}. \tag{5}$$

In Eq. (5), terms t^{pos} and f^{pos} indicates true positives and false positives," respectively.

4 Hand gesture recognition by adaptive region-based active contour

4.1 Image pre-processing

The proposed model uses median filtering (Zhu and Huang 2012) for preprocessing the hand gesture images, which is the significant pre-processing approach that aims to eliminate the irrelevant noise from the images. It also recognizes or improves the edges in images. In general, it is a non-linear filter for substituting the noisy and neighbor pixels of images. It is sorted by using the gray level of the image. The pre-processed images $\mathrm{Im}^{MF}(pre)$ using median filtering are given in Eq. (6):

$$\mathrm{Im}^{ip(MF)}(a,b) = MF\left\{\mathrm{Im}^{ip}(a-v, b-w)v, w \in Q\right\}. \tag{6}$$

Here, the term Q denotes the 2-D mask, where the center value of the mask is referred to as $\mathrm{Im}^{ip}(v,w)$, and the original image is considered as $\mathrm{Im}^{ip}(a,b)$. This median filter offers better performance than other filtering approaches. Finally, the preprocessed images using median filtering are given as $\mathrm{Im}^{ip(MF)}$.

4.2 Adaptive region-based active contour segmentation

The preprocessed images using median filtering $\mathrm{Im}^{ip(MF)}$ are forwarded to the segmentation phase, where a new ARAC approach is utilized with the help of the OSV-BSO algorithm. This new technique helps in getting suitable and accurate segmented results by considering the objective as the minimization of the variance and maximization of entropy among segmented and original images.

Region-based active contour (Pratondo et al. 2017): It is one of the efficient segmentation algorithms, which is also efficient for poorly defined boundaries. This model is implemented by Chan and Vese (Pratondo et al. 2017) that is applied to the matrix for getting the optimal solutions. The solutions converge faster, where a function $F\left(q_1, q_2, Q\right)$ is introduced by Chan and Vese as given in Eq. (7):

$$F(q_1, q_2, Q) = \varsigma \cdot Length(Q) + \vartheta \cdot Area(inside(Q)) + \lambda_1 \int_{inside(Q)} \left| \mathrm{Im}^{ip(MF)}(u, v) - q_1 \right|^2 dudv + \lambda_2 \int_{outside(Q)} \left| \mathrm{Im}^{ip(MF)}(u, v) - q_2 \right|^2 dudv. \quad (7)$$

Here, the energy function can be derived with the input images as the preprocessed images using median filtering $\mathrm{Im}^{ip(MF)}$, some of the constants are given as $\varsigma \geq 0$, $\vartheta \geq 0$ and $\lambda_1, \lambda_2 > 0$, the evolving curve is given as Q and the values of $\mathrm{Im}^{ip(MF)}$ inside and outside of Q are represented as q_1 and q_2, respectively. Further, the minimization problem can be derived in Eq. (8):

$$\inf_{q_1, q_2, Q} F(q_1, q_2, Q). \quad (8)$$

It can be addressed through the level set approach and implicit representation of curve Q is given as the zero level set of a Lipschitz function $\varphi(u, v, t)$, which is derived in Eq. (9):

$$Q(t) = \{(u, v) | \varphi(u, v, t) = 0\}. \quad (9)$$

The initial contour is given as $\varphi(u, v, 0) = \varphi_0(u, v)$ for $t = 0$, where an artificial time variable is termed as t. Further, a gradient flow is proposed by Chan and Vese as derived in Eq. (10):

$$\frac{\partial \varphi}{\partial t} = \delta_\varepsilon(\varphi) \left[\begin{array}{l} \varsigma \cdot div\left(\frac{\nabla \varphi}{|\nabla \varphi|} \right) - \vartheta - \lambda_1 \left(\mathrm{Im}^{ip(MF)} - q_1 \right)^2 \\ + \lambda_2 \left(\mathrm{Im}^{ip(MF)} - q_2 \right)^2 \end{array} \right] = 0 \ in \quad \Omega, \quad (10)$$

$$\frac{\delta_\varepsilon(\varphi)}{|\nabla \varphi|} \frac{\partial \varphi}{\vec{\eta}} = 0 \ on \ \Omega. \quad (11)$$

Here, the normal derivative of φ- at the boundary is denoted as $\frac{\partial \varphi}{\vec{\eta}}$, the exterior normal to the boundary is denoted as $\vec{\eta}$, a bounded open subset of $\mathfrak{R}^2$ with its $\partial \varphi$ boundary is given as Ω and the regularized Dirac function is termed as δ_ε. Moreover, Eq. (11) has several parameters, which have to be tuned appropriately.

To enhance the performance of segmentation, the parameters ς and ϑ, which are constants and a maximum number of iteration J of the region-based active contour are optimized using the OSV-BSO algorithm. The solution encoding of the designed model is given in Fig. 3.

The parameter ς and ϑ a maximum number of iteration J of ARAC are optimized using the OSV-BSO algorithm, where the bounding limits ς and ϑ are given as 0.1 to 0.8, respectively. The maximum number of iterations lies among [50, 500]. The segmented images through ARAC are given as $\mathrm{Im}^{ip(ARAC)}$.

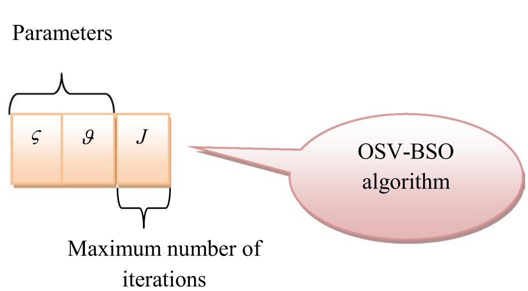

Fig. 3 Solution encoding of segmentation

4.3 Feature extraction

The segmented images using ARAC are subjected to the feature extraction stage to get the optimal features. The designed "hand gesture recognition model" uses the combination of feature extraction approaches like HOG (Kobayashi et al. 2008), ORB, and region props. The extracted features are given to the classification stage, where segmented images using region-based active contour $\mathrm{Im}^{ip(ARAC)}$ are given to all the techniques.

ORB (Aglave and Kolkure 2015): It improves the performance of features with rotation and orientation. It is one of the fast binary descriptors. It is resistant to noise, which works by returning the ORB points for the object that includes information about key points of ORB. The identification of ORB key points is performed from the input images. The extracted features using ORB are denoted as in_{fe}^{ORB}.

Region props (Yang et al. 2008): It is employed for extracting the significant features for deciding the objects in the binary image, which is called a pupil. It takes the input as $\mathrm{Im}^{ip(ARAC)}$. The "proposed hand gesture recognition "model uses some of the region props techniques like area, centroid, bounding box, orientation, eccentricity, and perimeter. This approach is used for measuring the properties of the image region.

Area: This property is used for detecting the area of the object, which means the number of pixels on the object.

Centroid: It is employed for getting the coordinates of the center of mass of the object for determining the center location. It is also denoted by rows and columns.

Bounding box: It is a simple computational geometry technique for computing the homeomorphisms among lattices and shapes. It is invariant to translation, scaling, robust and rotation against noisy shape boundaries. It considers the five steps for splitting the bounding box, where the bounding box is initially computed of a pixel set. The set of vertical slices is subdivided. The bounding box of every resultant pixel group is determined. The horizontal slices of every resultant pixel set are calculated.

Eccentricity: It is the metric of aspect ratio, which is the ratio of "the length of the major axis to the length of the minor axis." this method can be computed with minimum bounding rectangle approach or principal axes approach.

5 Proposed OSV-BSO for enhancing segmentation and classification

5.1 Proposed OSV-BSO

The “proposed hand gesture recognition “model develops a new algorithm called OSV-BSO for the optimization of the parameters ς and ϑ, and a maximum number of iteration J in the ARAC segmentation stage and weights of PNN WE_{NW} that aims to increase the convergence rate. This new variant of the BSO (Chen et al. 2018) algorithm is inspired by the foraging behavior of beetles along with the formulation of the PSO algorithm.

A new OSV-BSO algorithm develops a new velocity updating equations by assigning the values for cn_1 and cn_2, which is given here. A new velocity updating equation based on best solutions is given in Eq. (12):

$$Ve_{iv}^{j+1} = \omega Ve_{iv}^{j} + cn_1\left(Z_{iv}^{j} - Y_{iv}^{j}\right) \tag{12}$$

Here, leadership position is given as Z_{iv}^{j} and current position of beetles is specified as Y_{iv}^{j} and cn_1 is formulated in Eq. (13):

$$cn_1 = \frac{FF_{best}}{mean(FF)}. \tag{13}$$

In Eq. (13), the term FF_{best} is the leading fitness, and the mean of the fitness function is denoted as $mean(FF)$. Similarly, the velocity updating equation based on the worst fitness solution is equated in Eq. (14):

$$Ve_{iv}^{j+1} = \omega Ve_{iv}^{j} + cn_2\left(Z_{gv}^{j} - Y_{gv}^{j}\right). \tag{14}$$

In Eq. (14), the terms Z_{gv}^{j} and Y_{gv}^{j} indicate the worst and current position of beetles and cn_2 are given in Eq. (15):

$$cn_2 = \frac{FF_{worst}}{mean(FF)}. \tag{15}$$

Here, FF_{worst} is the worst leader function. The proposed OSV-BSO algorithm works by the random value. If $rand < 0.5$, then the position updating is performed with new velocity updating equation using best fitness solutions; or else the position updating is done by new velocity updating equation with worst fitness solutions. The flowchart of the proposed OSV-BSO algorithm is given in Fig. 4.

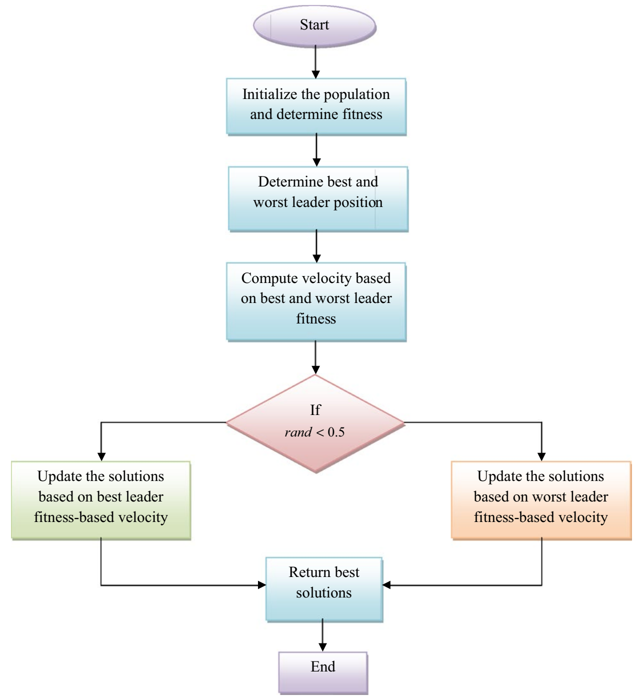

Fig. 4 Flowchart of the proposed OSV-BSO algorithm

5.2 Optimized PNN-based recognition

The “proposed hand gesture recognition “model use PNN (Tuccitto et al. 2019) for the classification of hand gestures by using the relevant features in^{Fe} attained from the feature extraction stage. It is insensitive to outliers and faster than the perceptron network. PNN provides diverse features like better avoidance of local minima problems, offers more accuracy, better prediction, faster training, and guarantees improved coverage for an optimal classifier “when the size of the training set” is increased. PNN is provided an accurate classification with four diverse polymers like "Poly Styrene (PS), Poly Methyl Meth Acrylate (PMMA), Polyalpha Methyl Styrene (PaMS), and Poly Ethylene Terephthalate-Co-Isophthalate (PETi)". Here, PNN includes multiple sub-networks, where each sub-network is considered as a pdf estimator for each class. There are four layers in PNN, in which the input layer includes two inputs such as kurtosis and skewness, the second layer has 300 neurons, the third layer offers an average operation of the results from the second layer at each class and the fourth layer consists of five neurons.

The proposed PNN architecture is given in Fig. 5.

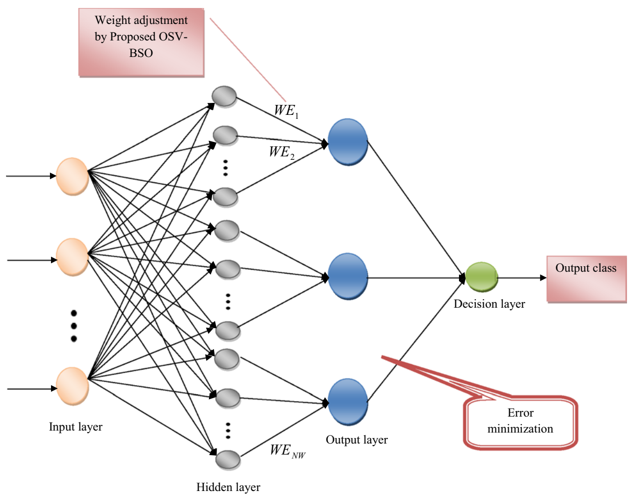

Fig. 5 Proposed optimized PNN for hand gesture recognition

6 Results and discussions

6.1 Experimental setup

The "proposed hand gesture recognition" in HCI was implemented in MATLAB 2020a, and the experimental analysis was carried out. The proposed model was executed by considering the maximum number of population as 10 and the maximum number of iterations as 25. Here, the performance of the proposed model was analyzed over the existing models concerning performance measures. It was compared with diverse optimization techniques like GWO (Sreedharan et al. 2018), PSO (Bonyadi and Michalewicz 2016), WOA (Mirjalili and Lewis 2016) and BSO (Chen et al. 2018) and traditional classifiers such as DT (Tsang et al. 2011), SVM (Wu and Yang 2015), NN (Wu et al. 2016) and PNN (Tuccitto et al. 2019).

6.2 Performance measures

Performance measurement is the process of collecting, analyzing, and/or reporting information regarding the performance of an individual, group, organization, system, or component. Definitions of performance measurement tend to be predicated upon an assumption about why the performance is being measured (https://en.wikipedia.org/wiki/Performance_measurement).

6.3 Performance analysis on different classifiers

The performance analysis of the suggested model with diverse classifiers by varying learning percentages is given for dataset 1 and dataset in Figs. 6 and 7, respectively. The performance analysis for dataset 1 is validated here. The accuracy of the proposed OSV-BSO-based PNN is 2%, 1%, 3%, and 1% better than DT, SVM, NN, and PNN, respectively at 35%.

Likewise, the experimentation results of dataset 2 are also analyzed here. The accuracy of the OSV-BSO-PNN is 2% advanced than DT, 3% advanced than SVM and NN, and 1% advanced than PNN at 45%. Hence, the "proposed hand gesture recognition" model establishes better performance than conventional techniques with diverse performance metrics.

6.4 Overall performance analysis for dataset 1

The comparative analysis of the suggested hand gesture recognition model for dataset 1 is represented in terms of diverse optimization algorithms and classifiers as given in Tables 1 and 2, respectively. The accuracy of the proposed OSV-BSO-PNN is 1%, 0.68%, 0.12%, and 1.3% improved than GWO-PNN, PSO-PNN, WOA-PNN, and BSO-PNN, respectively. Thus, the overall performance analysis of the "proposed hand gesture recognition model" using OSV-BSO-PNN for dataset 1 is improved than other traditional approaches.

6.5 Overall performance analysis for dataset 2

The performance of the "proposed hand gesture recognition model" for dataset 2 is analyzed with conventional optimization and machine learning approaches, which are given in Tables 3 and 4, respectively. The specificity of the developed OSV-BSO-based PNN is 0.6%, 0.69%, 0.7%, 0.75%, 0.68%, 0.8%, 0.66% and 0.8% improved than GWO-PNN, PSO-PNN, WOA-PNN, BSO-PNN, DT, SVM, NN and PNN, respectively. Therefore, the performance of the suggested "hand gesture recognition model" using OSV-BSO-based PNN is improved than other conventional approaches.

6.6 Running time analysis of OSV-BSO-PNN

The running time analysis of the proposed OSV-BSO-PNN model for two datasets is shown in Table 5. From the given Table 5, the OSV-BSO-PNN is attained 36.60%, 8.75%, 32.70%, and 48.60% reduced than GWO-PNN, PSO-PNN, WOA-PNN, and BSO-PNN, respectively. Therefore, it is proved that the proposed OSV-BSO-PNN is superior to the other conventional algorithms for all datasets.

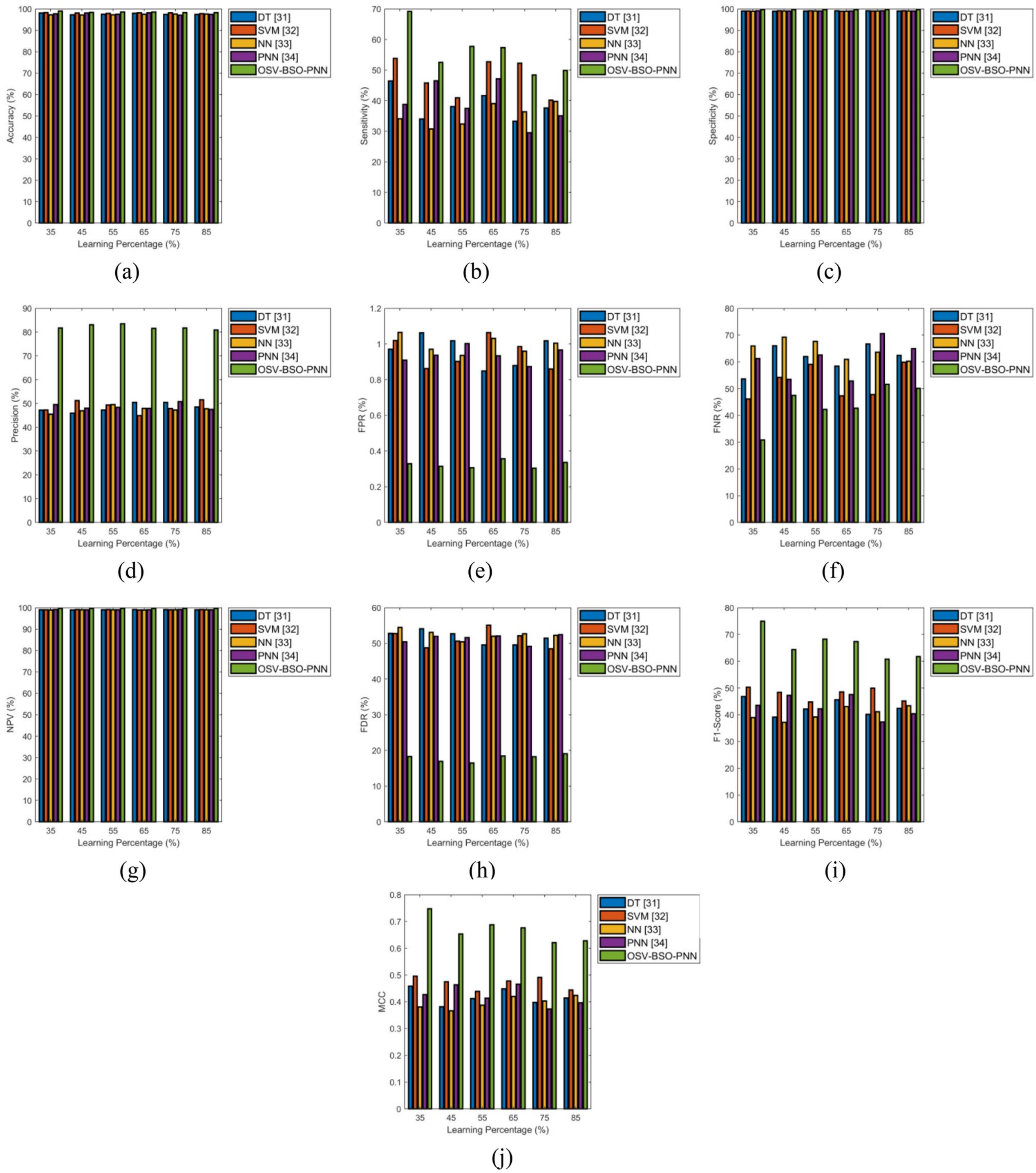

Fig. 6 Performance analysis of the proposed hand gesture recognition model with "proposed and conventional machine learning algorithms" for dataset 1 concerned with **a** accuracy, **b** sensitivity, **c** specificity, **d** precision, **e** FPR, **f** FNR, **g** NPV, **h** FDR, **i** F1-score and **j** MCC

7 Conclusion

This paper has planned to implement an intelligent hand gesture recognition model for HCI. The hand gestures from different benchmark sources were collected, and further, the pre-processing was offered through median filtering. Further, the ARAC method was utilized with the meta-heuristic algorithm for getting superior segmentation

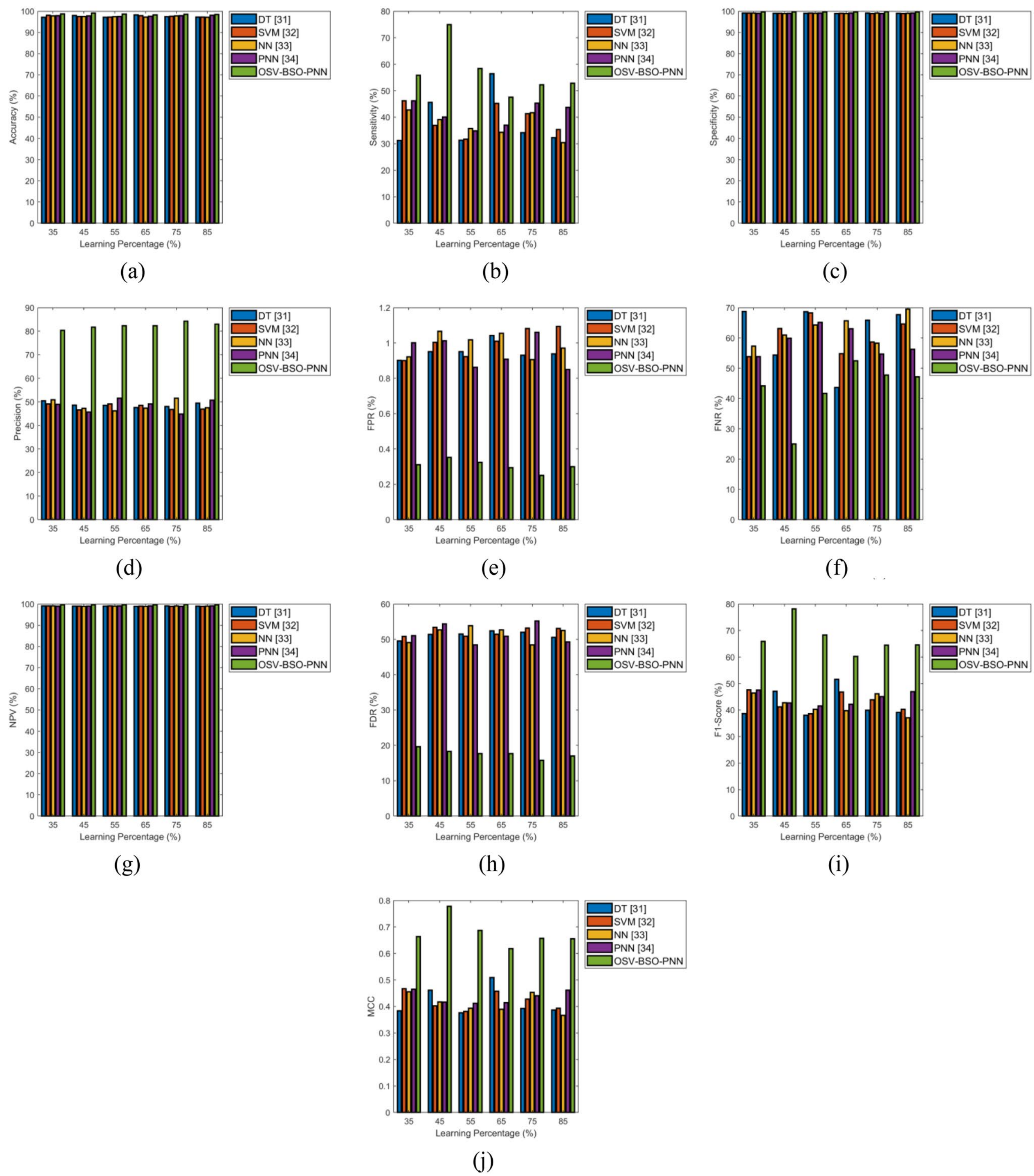

Fig. 7 Performance analysis of the proposed hand gesture recognition model with "proposed and conventional machine learning algorithms" for dataset 2 concerned with **a** accuracy, **b** sensitivity, **c** specificity, **d** precision, **e** FPR, **f** FNR, **g** NPV, **h** FDR, **i** F1-score and **j** MCC

results. Feature extraction was the next step, where features like HOG, ORB, and Regionprops techniques, which were dimensionality reduced by PCA. With these relevant features, the recognition phase was performed the optimized PNN through the OSV-BSO algorithm. Through the performance analysis, the accuracy of the proposed OSV-BSO-based PNN was 0.7%, 1.3%, 0.48%, 1.4%, 1.1%, 0.96%, 0.78%, and 0.6% superior to GWO-PNN,

Table 1 comparative analysis of the designed and traditional "heuristic-based hand gesture recognition" model

Measures	GWO-PNN (Sreedharan et al. 2018)	PSO-PNN (Bonyadi and Michalewicz 2016)	WOA-PNN (Mirjalili and Lewis 2016)	BSO-PNN (Chen et al. 2018)	OSV-BSO-PNN
Accuracy	0.97219	0.97622	0.98166	0.97012	0.98293
Sensitivity	0.33228	0.39135	0.5026	0.29954	0.48394
Specificity	0.98939	0.99069	0.99057	0.98968	0.99696
Precision	0.45727	0.50993	0.49775	0.45833	0.81745
FPR	0.010605	0.009306	0.009432	0.010323	0.003039
FNR	0.66772	0.60865	0.4974	0.70046	0.51606
NPV	0.98939	0.99069	0.99057	0.98968	0.99696
FDR	0.54273	0.49007	0.50225	0.54167	0.18255
F1-score	0.38488	0.44284	0.50016	0.3623	0.60796
MCC	0.37597	0.43485	0.49083	0.35596	0.62146

Table 2 Comparative analysis of the designed and traditional classifiers-based hand gesture recognition model

Measures	DT (Tsang et al. 2011)	SVM (Wu and Yang 2015)	NN (Wu et al. 2016)	PNN (Tuccitto et al. 2019)	OSV-BSO-PNN
Accuracy	0.97401	0.98218	0.97589	0.97057	0.98293
Sensitivity	0.33333	0.52232	0.36359	0.29485	0.48394
Specificity	0.99121	0.99015	0.9904	0.99126	0.99696
Precision	0.50447	0.47859	0.47298	0.50826	0.81745
FPR	0.008789	0.009854	0.0096	0.008735	0.003039
FNR	0.66667	0.47768	0.63641	0.70515	0.51606
NPV	0.99121	0.99015	0.9904	0.99126	0.99696
FDR	0.49553	0.52141	0.52702	0.49174	0.18255
F1-score	0.40142	0.4995	0.41113	0.3732	0.60796
MCC	0.39745	0.49093	0.40264	0.37326	0.62146

Table 3 Comparative analysis of the proposed and traditional optimization algorithm-based "hand gesture recognition" model

Measures	GWO-PNN (Sreedharan et al. 2018)	PSO-PNN (Bonyadi and Michalewicz 2016)	WOA-PNN (Mirjalili and Lewis 2016)	BSO-PNN (Chen et al. 2018)	OSV-BSO-PNN
Accuracy	0.97804	0.97253	0.98089	0.97126	0.98565
Sensitivity	0.40271	0.31272	0.46346	0.31126	0.52261
Specificity	0.99147	0.99062	0.99065	0.99	0.99749
Precision	0.52439	0.47758	0.48329	0.46922	0.84202
FPR	0.008528	0.009379	0.009349	0.009998	0.002508
FNR	0.59729	0.68728	0.53654	0.68874	0.47739
NPV	0.99147	0.99062	0.99065	0.99	0.99749
FDR	0.47561	0.52242	0.51671	0.53078	0.15798
F1-score	0.45556	0.37796	0.47317	0.37426	0.64493
MCC	0.44859	0.37311	0.46354	0.36813	0.657

Table 4 Comparative analysis of the proposed and traditional classifiers-based hand gesture recognition model

Measures	DT (Tsang et al. 2011)	SVM (Wu and Yang 2015)	NN (Wu et al. 2016)	PNN (Tuccitto et al. 2019)	OSV-BSO-PNN
Accuracy	0.97475	0.97621	0.97796	0.97939	0.98565
Sensitivity	0.34139	0.41328	0.4171	0.45321	0.52261
Specificity	0.99069	0.98918	0.99094	0.98939	0.99749
Precision	0.47995	0.46806	0.5159	0.44808	0.84202
FPR	0.00931	0.010821	0.009061	0.010609	0.002508
FNR	0.65861	0.58672	0.5829	0.54679	0.47739
NPV	0.99069	0.98918	0.99094	0.98939	0.99749
FDR	0.52005	0.53194	0.4841	0.55192	0.15798
F1-score	0.39898	0.43897	0.46127	0.45063	0.64493
MCC	0.39232	0.42773	0.4528	0.44014	0.657

Table 5 Comparative analysis of the proposed and traditional optimization algorithm-based "hand gesture recognition" model

Algorithms	GWO-PNN (Sreedharan et al. 2018)	PSO-PNN (Bonyadi and Michalewicz 2016)	WOA-PNN (Mirjalili and Lewis 2016)	BSO-PNN (Chen et al. 2018)	OSV-BSO-PNN
Dataset1	1.6761	1.3344	495.31	1.8234	1.227
Dataset2	1.4715	1.1935	44.566	1.1829	1.1136

PSO-PNN, WOA-PNN, BSO-PNN, DT, SVM, NN, and PNN, respectively. Finally, the experimental results have verified that the designed model has shown better performance with other comparative methods. Some of the failure cases in the proposed model are rotation problem that arises when the hand region is rotated in any direction in the scene. Translation problem arises when the variation of hand positions in different images also leads to an erroneous representation of the features. Accordingly, the scale problem appears when the hand poses have different sizes in the gesture image. In the future, the proposed model can be extended for processing larger-scale datasets by including the hand gesture recognition technology to achieve a very effective shape of the hand, and thus allows the extraction of robust and effective features and also aims to extend this mechanism to a range of disabled users.

Data availability statement The data underlying this article are available in "https://www.kaggle.com/ayuraj/american-sign-language-dataset:Access Date: 2021-01-11". "https://www.kaggle.com/muhammadkhalid/sign-language-for-alphabets: Access Date: 2021-01-11" and "https://www.kaggle.com/muhammadkhalid/sign-language-for-numbers/vaersion/1: Access Date: 2021-01-11".

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